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Manipulation of aqueous droplets in oil using contactless ultrasound

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ABSTRACT

Acoustic droplet manipulation is widely used in biochemical analysis due to its excellent biocompatibility and non-contact operation. However, conventional methods performed in air suffer from aerosol-induced sample cross-contamination. To address this limitation, we propose a quad-focused acoustic tweezer (QFAT) system composed of four ultrasonic transducers orthogonally arranged with a 45° inclination angle. The interference of multiple acoustic beams in the oil phase generates a low-pressure acoustic potential well, and the resulting acoustic radiation force enables non-contact droplet manipulation. Benefiting from the intrinsic isolation of aqueous droplets by oil phase, the system significantly reduces the risk of aerosol contamination. The QFAT enables stable droplet transport and, when integrated with an automated three-axis translation stage, guides droplets along complex trajectories such as circles and hexagons with high precision. In addition, the system supports parallel collective transport of multiple droplets for orderly multi-step chemical reactions, demonstrating strong potential for parallel biochemical manipulation in oil-phase environments. This work provides a robust and highly flexible non-contact droplet manipulation method with significantly reduced contamination risks for droplet-based biochemical analysis and synthesis.

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14 November 2026 07:57:31

Droplet manipulation enables precise control of discrete fluid units and their biochemical reaction via external fields, surface patterning, or gradient structures.[1,2] Owing to its low reagent consumption and high-throughput capability, it has become an important tool in biochemical analysis, with applications in protein analysis, polymerase chain reaction (PCR) assays, DNA sequencing, etc. Notably, droplet manipulation on superhydrophobic surfaces (DMSS) provides an open-surface platform that overcomes the fixed transport paths of conventional closed microchannels. Consequently, this approach facilitates highly programmable biochemical processes and improved reagent utilization.[2,3] By integrating external physical fields, including acoustic, electric, magnetic, and optical fields, DMSS can achieve precise two-dimensional droplet transport, fusion, splitting, and mixing, forming highly flexible and programmable "lab-on-a-surface" systems.[1,3–6]

Among these driving approaches, acoustic droplet manipulation (ADM) is particularly attractive because of its excellent biocompatibility and non-contact nature, requiring neither chemical modification nor additional materials.[3,7] However, current ADM systems operate almost exclusively in air which causes severe challenges in biochemical applications.[3,5,8–11] For example, evaporation during ultrasonic actuation and acoustothermal heating alters reagent concentrations in sensitive reactions such as PCR.[12,13] More critically, airborne aerosols generated during manipulation can lead to serious sample cross-contamination and false-positive PCR results.[14,15] Encapsulating droplets in a continuous oil phase can effectively isolate them from air, but the much smaller acoustic impedance mismatch at the droplet/oil interface greatly weakens acoustic scattering and radiation forces.[16] As a result, conventional ADM strategies designed for air

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cannot provide sufficient force to overcome viscous drag and substrate adhesion in oil, making efficient droplet transport difficult.

Several recent studies have explored oil-phase ADM to physically isolate samples and reduce aerosol contamination. Lin *et al.* developed a single-beam acoustic tweezer for trapping and selectively sampling droplets in oil.[17] However, the limited lateral radiation force of the narrow single-beam topology restricts free 2D droplet transport and parallel collective manipulation. Other oil-phase ADM strategies rely largely on physical confinement, such as closed microchannels[18] or pre-patterned static microwells,[19] limiting droplets to one-dimensional translation or fixed-point trapping, and thus reducing manipulation flexibility. Recently, Yiannacou *et al.* demonstrated 2D planar acoustic manipulation in oil through resonant-frequency scanning.[20] In this approach, changing the frequency shifts the standing-wave nodes, causing droplets to migrate between discrete acoustic potential wells, which is not suited to continuous, smooth, and high-precision translation along complex trajectories. Accordingly, highly programmable 2D ultrasonic manipulation of droplets in oil phase remains challenging.

To address this issue, we propose a quad-focused acoustic tweezer (QFAT) for oil-phase droplet manipulation. The system employs four ultrasonic transducers orthogonally arranged with a 45° tilt angle to generate a controllable acoustic potential well in the oil phase. Mounted on a three-axis translation stage, the system enables highly programmable 2D ultrasonic manipulation of droplets in oil phase, fundamentally reducing the risk of aerosol cross-contamination. Finite-element simulations were used to optimize the QFAT design and were validated experimentally. The optimized system achieved stable droplet transport and high-precision manipulation along complex trajectories, including circles and regular hexagons. In addition, QFAT enabled parallel collective transport and fusion of multiple droplets in oil. As a proof of concept, CH_3COOH and KOH droplets were sequentially merged with litmus reagent droplets, producing visible color changes and demonstrating the potential of QFAT for parallel biochemical analysis.

As shown in Fig. 1(a), the QFAT ultrasonic assembly consists of four cylindrical ultrasonic transducers (diameter: 21 mm) arranged orthogonally with a 45° inclination angle in a centrosymmetric configuration (see details in Fig. S1). A sinusoidal electrical signal (950 kHz, initial amplitude of 1.4 V peak-to-peak) is amplified to 27.9 V peak-to-peak and applied to the transducers. All four transducers are completely wired in parallel with identical signal amplitude and phase. The oil phase is polydimethylsiloxane with a viscosity of 0.65 cSt (see detailed information in the supplementary material). The interference and superposition of the four acoustic beams in the oil phase produce a stable acoustic field distribution. As illustrated in Fig. 1(a), red and light gray denote high- and low-pressure regions, respectively. Periodically distributed high-pressure nodes form a low-pressure acoustic potential well with a clear boundary near the center of the chamber bottom.

As shown in Fig. 1(a), the droplet is initially located on a slightly left side of the central acoustic well, with a high-pressure region on its

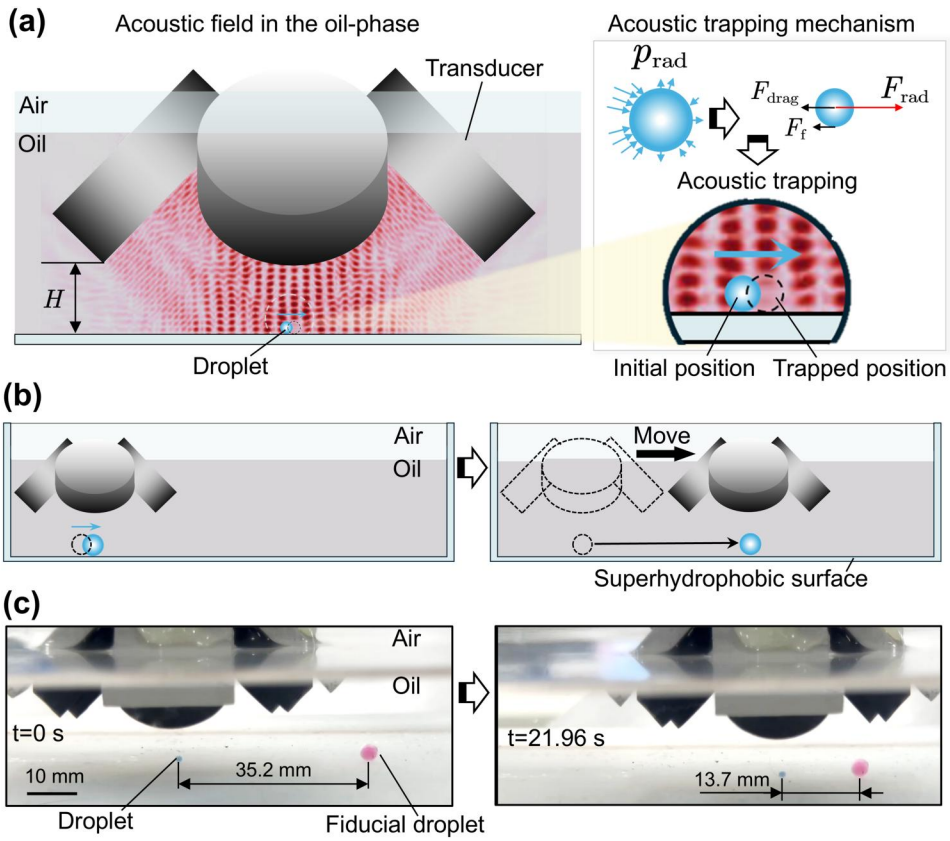


FIG. 1. Principle analysis and performance demonstration of quad-focused acoustic tweezer (QFAT) driven droplet manipulation within the continuous oil phase. (a) Schematic of the acoustic field distribution and droplet trapping mechanism. A distinct and stable low-pressure acoustic trap is formed at the bottom center at a QFAT elevation H of 7.60 mm. Here, p_{rad}, F_{rad}, F_{drag}, and F_f denote the acoustic radiation pressure, acoustic radiation force, viscous drag force, and interfacial friction force, respectively. (b) Targeted trapping and dynamic translation process of the droplet in the oil medium. (c) Visual demonstration of non-contact droplet manipulation within the oil system. The predefined displacement distance of the transducer assembly during this 21.96 s period is 21.74 mm.

14 November 2026 07:57:31

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left and the low-pressure well on its right. COMSOL simulations (see details in Sec. 1.5 of the supplementary material) indicate that the radiation pressure exerted from the left side of the droplet is significantly greater than that from the right side. This asymmetric pressure distribution generates a net acoustic radiation force directed toward the central acoustic well. When $\boldsymbol{F}_{\mathbf{rad}}$ exceeds the sum of friction force and drag force $\boldsymbol{F}_{\mathbf{f}} + \boldsymbol{F}_{\mathbf{drag}}$,[21] the droplet is driven back and stably trapped within the acoustic potential well. Here, $\boldsymbol{F}_{\mathbf{f}}$ and $\boldsymbol{F}_{\mathbf{drag}}$ represent the interfacial friction force from the superhydrophobic substrate and the viscous drag force from the continuous oil medium, respectively (see details in the supplementary material). When the QFAT translates to the right, the central acoustic well shifts synchronously. A droplet initially trapped in the well then lags behind and becomes displaced leftward, breaking the stress symmetry on its surface and re-generating a net acoustic radiation force $\boldsymbol{F}_{\mathbf{rad}}$ in the direction of QFAT motion. This force drives the droplet to follow the moving acoustic well and become stably re-trapped.[7] Such a continuous "lag-correction" mechanism enables free 2D droplet transport in the oil phase. A single localized low-pressure acoustic well can stably trap droplets with diameters comparable to the well size, typically on the order of half the acoustic wavelength.[22–24] As shown in Fig. 1(a), additional low-pressure wells are periodically formed within the interference acoustic field, and stable parallel manipulation of multiple droplets can be achieved under appropriate inter-droplet spacing,[7] which is demonstrated in Fig. 4.

Figure 1(b) illustrates the manipulation procedure of QFAT in the oil phase. First, a droplet is trapped within the low-pressure acoustic well through acoustic excitation. Subsequently, translation of the transducer assembly generates a non-contact pulling effect, enabling arbitrary two-dimensional droplet motion on the superhydrophobic substrate. As shown in Fig. 1(c) and Movie S1, a blue target droplet (diameter: 0.92 mm) and a red reference droplet were initially placed in the oil phase. After acoustic excitation was applied at $t = 0.00$ s, the blue droplet was rapidly trapped and then transported 21.49 mm toward the reference droplet within 21.96 s under contactless acoustic actuation.

The operating frequency of QFAT was fixed at 950 kHz, and the corresponding acoustic wavelength is approximately 0.92 mm (see the supplementary material for the details of material properties and numerical simulation). Since the trapping capability of a localized acoustic well is closely related to its physical width, the effective well size fundamentally determines the maximum trappable droplet size. To optimize manipulation stability and trapping range, parametric simulations were performed to identify the conditions producing the widest acoustic well. As shown in Fig. 2(b), the focal position coinciding with the chamber bottom is defined as the zero relative height ($\Delta H = 0$). Simulations were conducted over a range from -6λ to 6λ (see Figs. S6 and S7 in the supplementary material for complete results). Comparison of the acoustic field distributions revealed that the widest effective acoustic well was obtained at $\Delta H = -1.75\lambda$, corresponding to a transducer-to-substrate distance of approximately

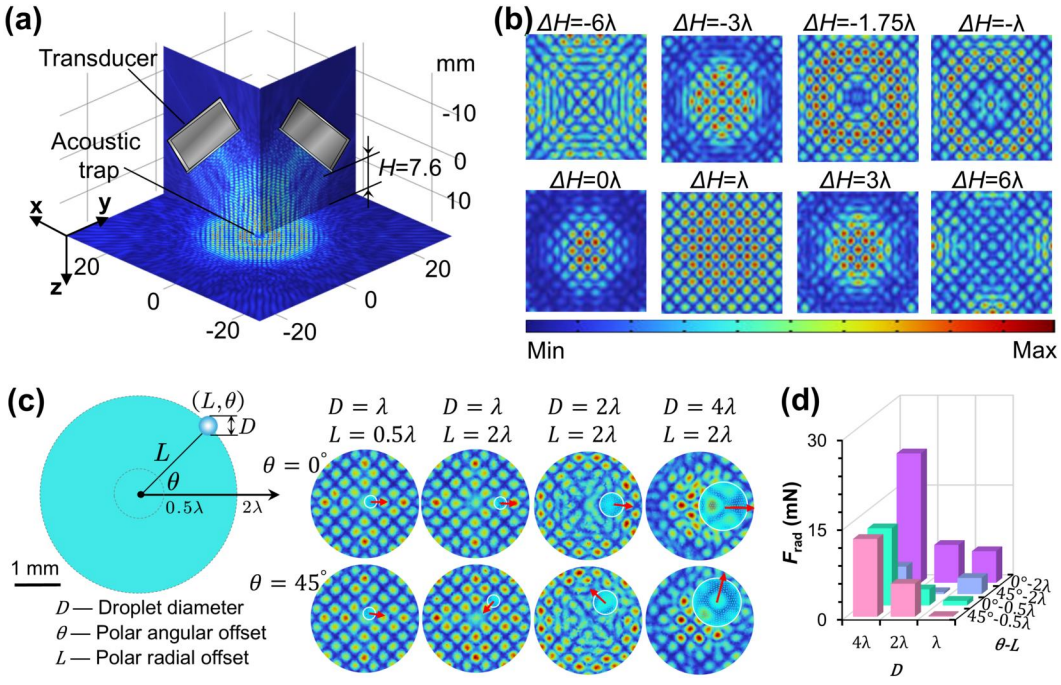


FIG. 2. Three-dimensional acoustic field distribution of the QFAT within the oil phase and the related parametric simulation results. (a) Three-dimensional acoustic field distribution at a QFAT elevation H of 7.60 mm. (b) Top view acoustic field distributions at the oil chamber bottom under varying QFAT elevations. A distinct and broad low-pressure acoustic potential well is formed at the bottom center at $\Delta H = -1.75\lambda$ representing the optimal manipulation conditions. (c) Effects of droplet size and spatial position on the exerted acoustic radiation force (indicated by the red arrow) under optimal conditions. (d) Quantitative comparison of the acoustic radiation forces exerted on droplets across different sizes and spatial deviations. Here, ΔH is the relative height of the QFAT, λ represents the acoustic wavelength, D is the droplet diameter, and L and θ denote the polar radial offset and polar angular offset of the droplet, respectively.

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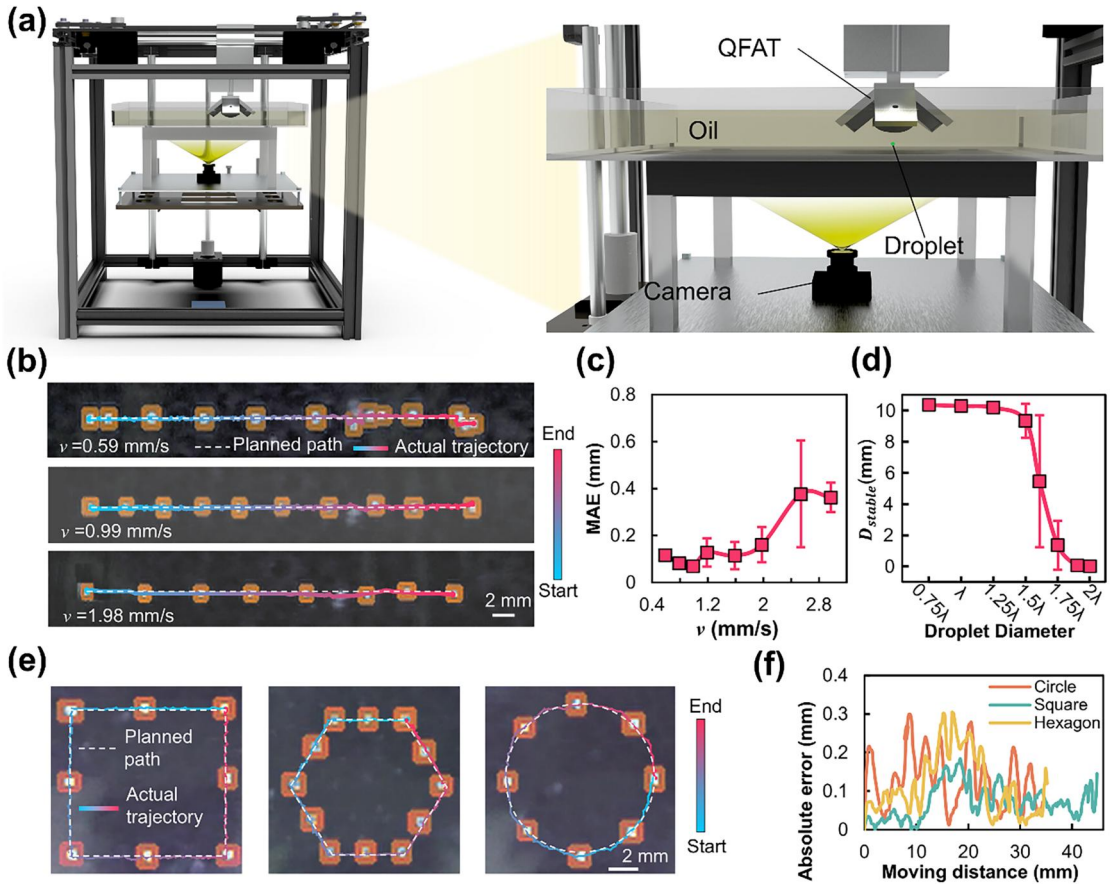


FIG. 3. QFAT-based automated oil-phase droplet manipulation platform and quantitative demonstration of manipulation accuracy for aqueous droplets within the oil system. (a) Hardware setup of the automated oil-phase droplet manipulation platform integrating a camera for machine vision. (b) QFAT driven long distance transportation of droplets within the oil phase at varying translation speeds (v). (c) Variation of the mean absolute error (MAE) with translation speed (v). (d) Dependence of stable moving distance (D_{stable}) on droplet diameter under a constant transducer displacement. (e) Non-contact manipulation of droplets tracking three complex trajectories within the oil phase. (f) Absolute error distributions of droplets tracking the three complex trajectories.

7.60 mm. This condition was therefore selected as the optimal manipulation state. The corresponding three-dimensional acoustic field distribution is shown in Fig. 2(a), where a distinct low-pressure acoustic well is clearly observed at the geometric center.

The effects of droplet size and spatial deviation on the acoustic radiation force were further investigated numerically under the optimal condition, as shown in Fig. 2(c), where red arrows denote the force direction. Here, D is the droplet diameter, L is the distance from the chamber center, and θ is the deviation angle. For $D = \lambda$, the acoustic radiation force at different positions consistently points toward the nearest low-pressure well, indicating stable trapping of droplets with diameters of approximately 0.92 mm. In contrast, for larger droplets ($D = 2\lambda$ or 4λ), the force becomes strongly repulsive. When $L = 2\lambda$ for the larger droplets, the force vectors always point away from the center, causing droplets to escape the acoustic well. This agrees with previous studies by Marzo and coworkers,[23–25] which showed that larger droplets span multiple pressure nodes and antinodes, leading to conflicting acoustic forces and outward repulsion. Correspondingly, the regular periodic acoustic field observed at $D = \lambda$ becomes severely distorted for larger droplets, as shown in Fig. 2(c).

Our experiments further confirmed this mechanism: QFAT could stably trap only droplets with $D = \lambda$, whereas larger droplets ($D > \lambda$) were immediately repelled from the localized acoustic well after acoustic excitation. The corresponding experimental video is provided in Movie S2 of the supplementary material. Figure 2(d) further quantifies the combined effects of droplet size and spatial deviation on the acoustic radiation force magnitude, showing that $\boldsymbol{F}_{\mathbf{rad}}$ increases significantly with droplet diameter.[26–28]

Based on QFAT, we developed an automated oil-phase droplet manipulation platform, as illustrated in Fig. 3(a) (see details in Figs. S2 and S3). The transducers are fully immersed in the oil phase and mounted on a motor-driven support, enabling free two-dimensional motion of QFAT and thus high-flexibility droplet manipulation. A superhydrophobic substrate at the chamber bottom minimizes solid–liquid adhesion and allows droplets to move with low resistance.[29,30] Meanwhile, a bottom-mounted camera combined with a pre-trained NanoDet network provides

14 November 2026 07:57:31

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real-time droplet localization for visual-feedback control. The performance of droplet recognition is shown in Fig. S4.

Under the optimal manipulation condition ($H = 7.60$ mm), droplets with $D = \lambda(0.92\,\text{mm})$ were used for experiments. As shown in Fig. 3(b) and Movie S3, the system achieved stable transport of droplets over 35 mm at different speeds in the oil phase. The white droplet in the images is outlined by the orange recognition frame of the vision system. Figure 3(c) shows that the mean absolute error (MAE) increases nearly exponentially with transport speed.[28] When the speed exceeded approximately 3.5 mm/s, the acoustic radiation force became insufficient to sustain stable trapping, causing droplet escape.[31] At an optimal speed of 0.99 mm/s, the MAE was as low as 0.07 mm, demonstrating the high manipulation accuracy of QFAT in the oil phase.

Furthermore, we investigated the dependence of the stable moving distance (D_{stable}) on droplet diameter, as shown in Movie S7 and Fig. 3(d). Specifically, at a constant QFAT translation speed of 0.99 mm/s and a target displacement of 10 mm, we recorded the distance droplets could stably follow the directional movement (D_{stable}). Small droplets ($D \leq 1.25\lambda$) were reliably trapped and transported over the entire distance ($D_{stable} \approx 10$ mm). Conversely, large droplets ($D \geq 2\lambda$) experienced substantial repulsion and are immediately expelled from the trap center upon QFAT activation, preventing any stable transport ($D_{stable} \approx 0$ mm). For medium-sized droplets in the critical transition zone ($1.5\lambda \leq D \leq 1.75\lambda$), they can be marginally trapped upon QFAT activation and stably follow the QFAT for a certain distance. However, the additional viscous drag force (F_{drag}) generated during translation significantly increases the total resistance. This dynamic resistance breaks the force balance ($F_{rad} < F_f + F_{drag}$), causing these droplets to drift and escape after a certain distance ($0 < D_{stable} < 10$ mm). Note that while our system stably traps small droplets ($D \leq 1.25\lambda$), a theoretical lower size limit exists for subwavelength droplets. When the droplet radius $R \ll \lambda$ (in the Rayleigh regime),[32] the acoustic radiation force scales with the droplet volume ($F_{rad} \propto R^3$), whereas the linear resistive forces scale with the radius ($F_{drag} + F_f \propto R$) (see details in the supplementary material). As the droplet size decreases, this driving force decays much faster than the total resistance. Therefore, critically small droplets cannot be deterministically trapped.

To further improve manipulation precision, the system integrates a visual-feedback closed-loop control algorithm. By extracting droplet coordinates in real time and dynamically compensating motor motion, the system achieves automated high-precision tracking of arbitrary trajectories (see Fig. S5). As shown in Fig. 3(e), QFAT enabled droplets to accurately follow square, circular, and hexagonal trajectories, with MAEs of 0.112, 0.137, and 0.142 mm, respectively. Figure 3(f) shows that the instantaneous tracking error remained below 0.3 mm throughout the motion process. The trajectory tracking results and video are provided in Fig. S8 and Movie S4 of the supplementary material.

As discussed above, QFAT generates multiple acoustic wells in the oil phase, enabling parallel collective transport of multiple droplets. Figure 4(a) demonstrates contactless parallel transport of two droplets with diameters of 0.94 and 0.88 mm, while Fig. 4(b) shows simultaneous manipulation of three droplets with diameters of 0.76, 0.78, and 0.72 mm (see Movie S5). Note that during the simultaneous manipulation of multiple droplets, interactions between adjacent droplets inevitably occur. As shown in Fig. 4(b), three droplets are

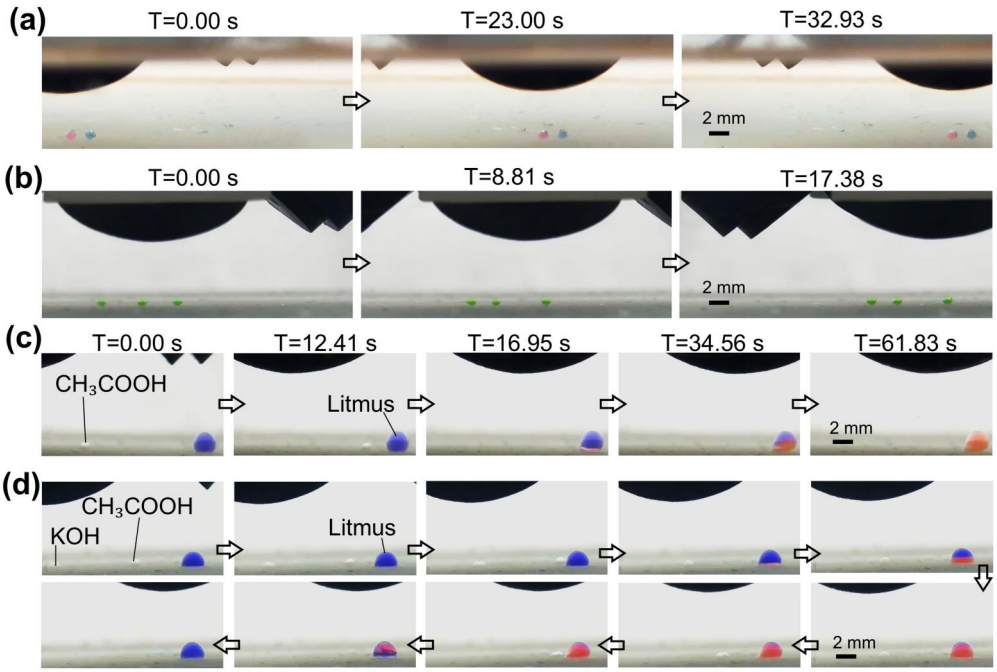


FIG. 4. Demonstration of the acoustic tweezer system in biochemical applications. (a) Parallel collective transport of two droplets within the oil phase. (b) Parallel collective transport of three smaller droplets. (c) Non-contact coalescence of an acetic acid droplet with a litmus solution within the oil phase. (d) Parallel collective transport of KOH and acetic acid droplets for sequential multi-step reactions with a litmus solution within the oil phase.

14 November 2026 07:57:31

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initially equally spaced at T = 0.00 s. As the parallel manipulation by the QFAT proceeds, at T = 8.81 s, the two leftmost droplets move slightly closer relative to their initial equidistant state. However, uncontrolled coalescence did not happen, and all droplets were stably confined in their respective traps and successfully transported as a whole to the final destination at T = 17.38 s. These results verify that stable parallel manipulation of multiple droplets can be achieved under appropriate inter-droplet spacing using multiple acoustic wells and highlight the potential of QFAT for parallel biochemical manipulation.

Using this automated oil-phase droplet manipulation platform, we further demonstrated contactless droplet fusion reactions in the oil phase. As shown in Fig. 4(c), an acetic acid droplet was manipulated toward a litmus solution droplet. The two droplets fused at T = 16.95 s, and the litmus solution gradually changed color from purple to red, completing the reaction by T = 61.83 s. Building on the demonstrated capabilities of parallel transport and targeted fusion, we further achieved sequential multi-droplet reactions in the oil phase, as shown in Fig. 4(d) and Movie S6. QFAT first transported KOH (1 mol/L) and CH_3COOH (10%) droplets in parallel toward a litmus (1% w/v) droplet. The CH_3COOH droplet fused first, turning the litmus solution red, while the KOH droplet remained stationary. After the first reaction was completed, the KOH droplet was guided to fuse with the litmus solution, gradually changing its color from red to blue. This result confirms that the system can manipulate multiple droplets in parallel to execute sequential multi-step chemical reactions within the oil phase. The current litmus-based experiments serve primarily as proof-of-concept demonstrations. Further development is needed to extend the platform to applications such as PCR and drug screening. In particular, environmental control modules are currently lacking, including temperature regulation for PCR thermal cycling and live-cell experiments, as well as CO_2 control for cell culture. These capabilities should be incorporated in future studies.

In summary, this study systematically elucidates the mechanism of acoustic radiation force in the oil phase through theoretical analysis and numerical simulation, revealing the fundamental relationship between ultrasonic wavelength and droplet size for acoustic droplet manipulation. Based on this mechanism, we developed an automated acoustofluidic biochemical reaction platform using QFAT, enabling precise and controllable non-contact manipulation of aqueous droplets in a closed oil-phase environment. Benefiting from the intrinsic isolation effect of the oil phase, the proposed platform significantly reduces the risk of aerosol contamination, overcoming a major limitation of conventional droplet manipulation systems. This work provides a robust and flexible strategy for multiphase acoustofluidic biochemical reactions, with promising applications in PCR detection, micro/nanoscale biochemical synthesis, and parallel biochemical analysis and drug screening.

See the supplementary material for details on experimental materials and equipment, the construction of the quad-focused acoustic tweezers (QFAT) and preparation of superhydrophobic surfaces, numerical simulations of the acoustic field and theoretical force analysis, the machine vision-based closed-loop control algorithm, and the high-precision tracking performance of droplets across complex square, hexagonal, and circular trajectories.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tao Luo: Conceptualization (lead); Funding acquisition (lead); Methodology (lead); Resources (lead); Supervision (lead); Writing – original draft (lead); Writing – review & editing (lead). **Chunrong Su:** Data curation (lead); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Yiying Tang:** Data curation (lead); Methodology (equal); Software (lead); Validation (equal); Visualization (equal). **Xinran Cao:** Data curation (equal); Methodology (equal); Software (lead); Validation (equal); Visualization (equal). **Yiwei Zhang:** Methodology (equal); Software (equal); Validation (equal); Visualization (equal). **Yingying Dai:** Methodology (equal); Software (equal); Validation (equal); Visualization (equal). **Dongyang Chen:** Software (equal); Validation (equal); Visualization (equal). **Jinyang Le:** Software (equal); Validation (equal). **Yuanpeng Lin:** Software (equal); Validation (equal). **Yujie Zheng:** Software (equal); Validation (equal). **Wendi Gao:** Conceptualization (lead); Methodology (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead). **Zhijie Huan:** Conceptualization (lead); Methodology (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead). **Wei zhou:** Conceptualization (lead); Funding acquisition (lead); Methodology (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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